Investigation of Targeted Malicious Activity Against European Infrastructure

A recent surge in malicious activity has been detected targeting critical infrastructure and government agencies across Europe. Current intelligence identifies five primary threat groups as suspected originators.

In response, my team is spearheading a deep-dive investigation into Sea Turtle. Leveraging real-world intelligence via the **MISP platform**, we are focused on:

> **(1) Infrastructure Mapping:** Identifying command-and-control (C2) and delivery systems.
>
> **(2) Technical Analysis:** Evaluating Indicators of Compromise (IOCs) to determine actor TTPs.
>
> **(3) Risk Mitigation:** Constructing comprehensive threat profiles to provide local organizations with actionable defensive strategies.

Threat Actor Investigation and Profiling

Threat Actor: Sea Turtle
Source: MISP Galaxy Intelligence + OSINT Correlation
Analyst Context: Authorized Cybersecurity Research Lab
Platform: MISP Threat Sharing Platform
Classification: Strategic & Operational Threat Intelligence

Executive Summary

This investigation confirms the presence of a **high-confidence strategic cyber threat** linked to the **Sea Turtle threat actor**, identified through real-world intelligence correlation using MISP threat feeds and OSINT platforms. Sea Turtle is a **state-aligned cyber espionage group** known for long-term, stealthy operations focused on **infrastructure-level compromise rather than noisy malware activity**.

Unlike financially motivated cybercriminal groups, Sea Turtle operates with **strategic intent**, targeting government, military, telecom, and critical infrastructure sectors. Their primary technique—**DNS hijacking combined with Adversary-in-the-Middle (AiTM) operations**—allows them to silently intercept communications, harvest credentials, and manipulate network traffic without triggering traditional endpoint security controls. This makes their operations difficult to detect and particularly dangerous for organizations that rely mainly on endpoint protection.

MISP correlation confirms that this activity is **not isolated**. The actor demonstrates **persistent operational behavior**, infrastructure reuse, and long-term campaign continuity across multiple intelligence events. Their infrastructure includes **dedicated command-and-control nodes, rogue DNS infrastructure, MITM nodes, and actor-controlled name servers**, enabling deep network visibility and sustained access to targeted environments.

From a risk perspective, this threat model represents a **strategic exposure** rather than a single-incident risk. Successful operations by Sea Turtle can lead to:

a. Long-term data interception

b. Intelligence leakage

c. Credential compromise

d. Network trust erosion

e. Regulatory and compliance exposure

f. National and organizational security risk

This analysis demonstrates that **traditional security models are insufficient** against this class of actor. Sea Turtle does not depend on mass phishing, ransomware, or commodity malware; instead, they exploit **trust relationships in internet infrastructure itself**.

Key Security Implication for Leadership:
This threat validates the need to shift from reactive cybersecurity to **proactive, intelligence-driven defense**. Protection must extend beyond endpoints into **DNS security, network monitoring, infrastructure integrity, and threat intelligence integration**. **Application Layer Protocol**

MISP Investigation Methodology

Platform: MISP
Event ID: 1869
UUID: 4b79f6b1-f69c-483b-9d24-6e20039f5e96
Source Event: 187 (CUDESO, 2019-07-09)

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1. MISP Intelligence Search

Performed comprehensive queries within the MISP platform leveraging:

- Talos Intelligence
- MITRE ATT&CK Framework
- CK Framework

This step was conducted to validate indicators, identify related threat intelligence, and determine attribution patterns associated with the event.

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2. APT Profile Creation

Created a dedicated Advanced Persistent Threat (APT) profile within MISP:

- Event: 1869
- Profile Name: APT Profile – Sea Turtle DNS Hijacking Campaign

The profile consolidates relevant indicators, behavioral patterns, and contextual intelligence specific to the campaign.

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3. Applied Threat Actor Taxonomy

Applied appropriate threat actor taxonomy using:

- misp-galaxy:threat-actor
- Sea Turtle

Linked the threat actor to Cluster ID 81836 to establish attribution consistency and structured intelligence mapping.

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4. Indicator Correlation & IOC Merging

Merged all 15 verified Indicators of Compromise (IOCs) from Source Event 187 into Event 1869 using MISP's merge functionality.

This ensured:

- Consolidation of validated intelligence
- Elimination of duplicate entries
- Centralized event management

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5. TTP Documentation

Mapped and documented 9 MITRE ATT&CK Tactics, Techniques, and Procedures (TTPs).

TTPs were linked to Sea Turtle operations using MISP Galaxy relationships to ensure structured adversary profiling and analytical traceability.

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6. Intelligence Report Authoring

Produced a structured intelligence report within the MISP Event Reports module, including:

- Analyst findings
- Technical assessment
- Attribution rationale
- Supporting evidence and correlation analysis

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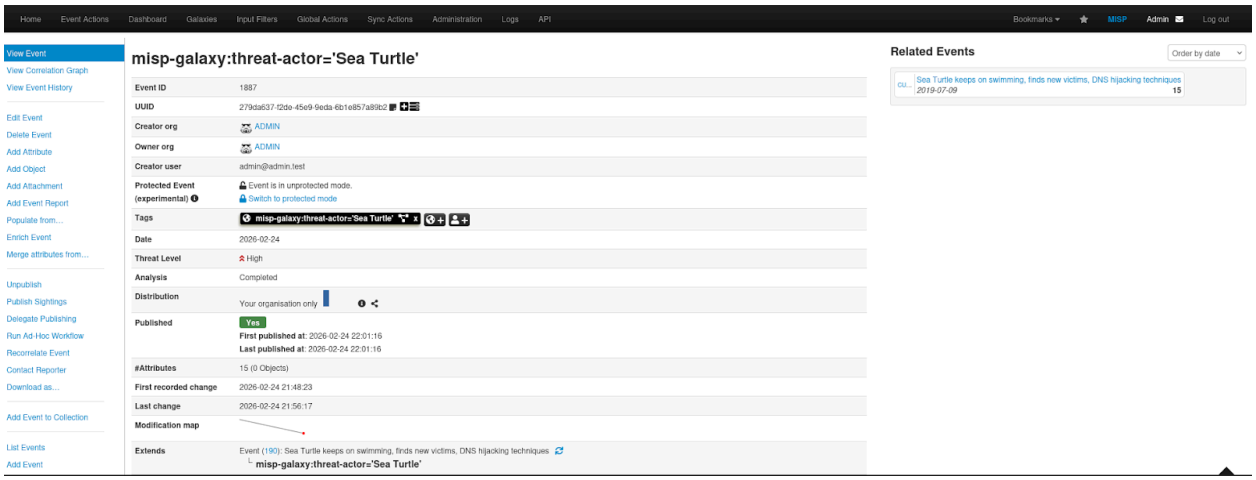
7. Publication & Export

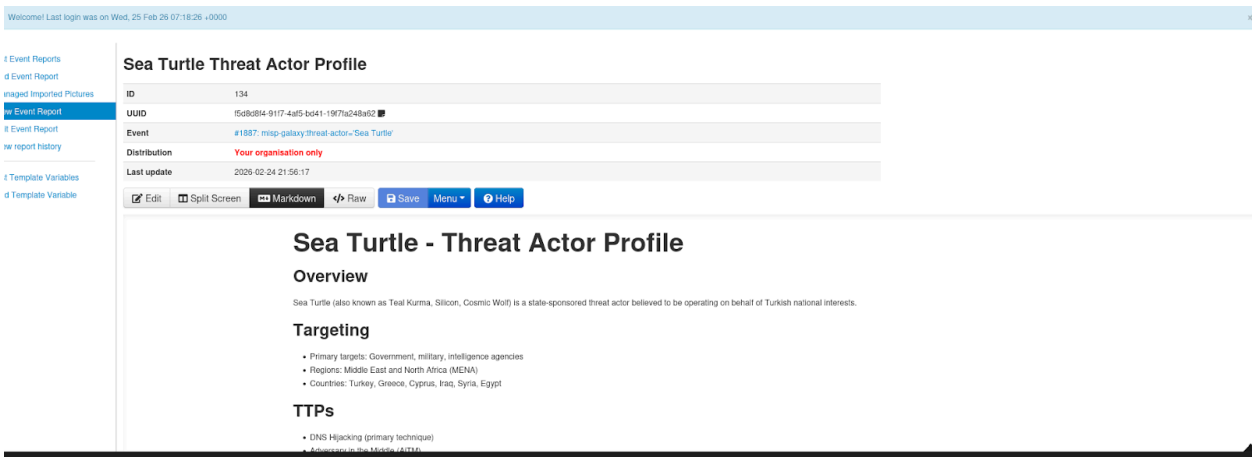
Published the finalized intelligence event and exported the data in STIX 2 format to facilitate:

- Information sharing
- Inter-organizational collaboration
- Threat intelligence dissemination across the community

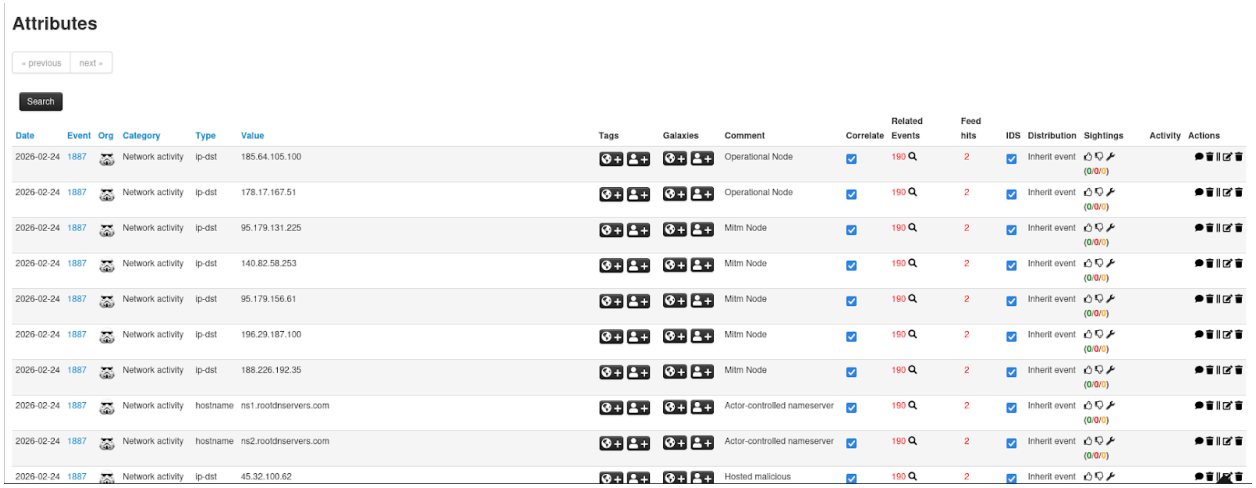
Strategic Security Positioning:
Organizations capable of integrating threat intelligence into real-time detection systems, enforcing DNS security governance, and deploying network-level monitoring will significantly reduce exposure to this type of actor. Those that do not will remain vulnerable to **silent, long-term compromise**.

Screen Short and pictures of the MISP threat profile which I built.





Attributes

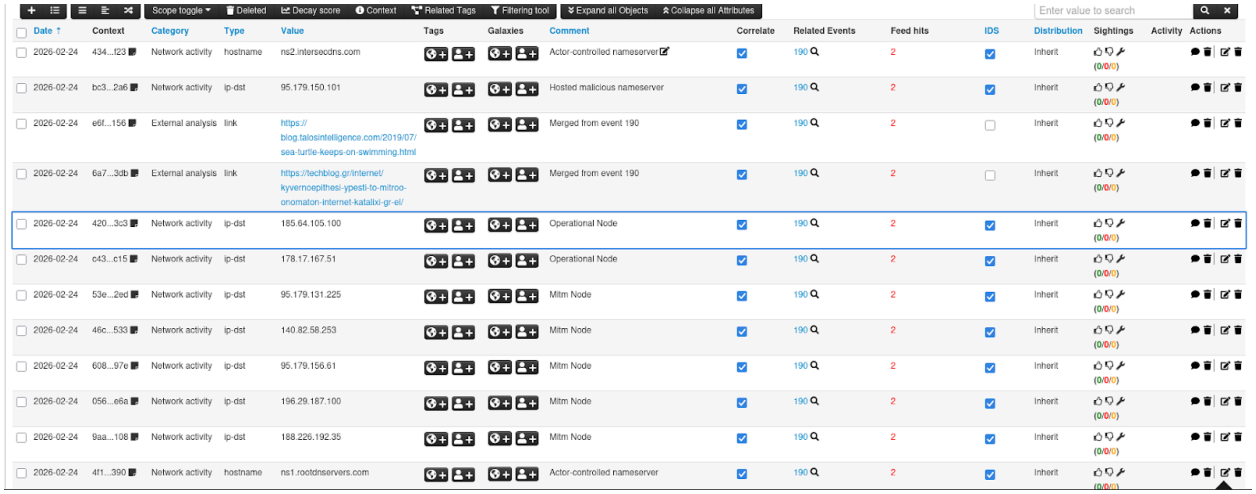


Attributes

« previous next »

Search

Date	Event	Org	Category	Type	Value	Tags	Galaxies	Comment	Correlate	Related Events	Feed hits	IDS	Distribution	Sightings	Activity	Actions
2026-02-24	1887		Network activity	ip-dst	185.64.105.100			Operational Node	✓	190	2	✓	Inherit event	(0/0/0)		
2026-02-24	1887		Network activity	ip-dst	178.17.167.51			Operational Node	✓	190	2	✓	Inherit event	(0/0/0)		
2026-02-24	1887		Network activity	ip-dst	95.179.131.225			Mitm Node	✓	190	2	✓	Inherit event	(0/0/0)		
2026-02-24	1887		Network activity	ip-dst	140.82.58.253			Mitm Node	✓	190	2	✓	Inherit event	(0/0/0)		
2026-02-24	1887		Network activity	ip-dst	95.179.156.61			Mitm Node	✓	190	2	✓	Inherit event	(0/0/0)		
2026-02-24	1887		Network activity	ip-dst	196.29.187.100			Mitm Node	✓	190	2	✓	Inherit event	(0/0/0)		
2026-02-24	1887		Network activity	ip-dst	188.226.192.35			Mitm Node	✓	190	2	✓	Inherit event	(0/0/0)		
2026-02-24	1887		Network activity	hostname	ns1.rootdnservers.com			Actor-controlled nameserver	✓	190	2	✓	Inherit event	(0/0/0)		
2026-02-24	1887		Network activity	hostname	ns2.rootdnservers.com			Actor-controlled nameserver	✓	190	2	✓	Inherit event	(0/0/0)		
2026-02-24	1887		Network activity	ip-dst	45.32.100.62			Hosted malicious	✓	190	2	✓	Inherit event			



Date	Context	Category	Type	Value	Tags	Galaxies	Comment	Correlate	Related Events	Feed hits	IDS	Distribution	Sightings	Activity	Actions
2026-02-24	434...f23	Network activity	hostname	ns2.interseccdns.com			Actor-controlled nameserver	✓	190	2	✓	Inherit	(0/0/0)		
2026-02-24	bc3...2a6	Network activity	ip-dst	95.179.150.101			Hosted malicious nameserver	✓	190	2	✓	Inherit	(0/0/0)		
2026-02-24	e6f...156	External analysis	link	https://blog.talosintelligence.com/2019/07/sea-turtle-keeps-on-swimming.html			Merged from event 190	✓	190	2		Inherit	(0/0/0)		
2026-02-24	6a7...3db	External analysis	link	https://techblog.gr/internet/kyvernoepithesi-ypesti-to-mitroo-onomaton-internet-katalixi-gr-el/			Merged from event 190	✓	190	2		Inherit	(0/0/0)		
2026-02-24	420...3c3	Network activity	ip-dst	185.64.105.100			Operational Node	✓	190	2	✓	Inherit	(0/0/0)		
2026-02-24	c43...c15	Network activity	ip-dst	178.17.167.51			Operational Node	✓	190	2	✓	Inherit	(0/0/0)		
2026-02-24	53e...2ed	Network activity	ip-dst	95.179.131.225			Mitm Node	✓	190	2	✓	Inherit	(0/0/0)		
2026-02-24	46c...533	Network activity	ip-dst	140.82.58.253			Mitm Node	✓	190	2	✓	Inherit	(0/0/0)		
2026-02-24	608...97e	Network activity	ip-dst	95.179.156.61			Mitm Node	✓	190	2	✓	Inherit	(0/0/0)		
2026-02-24	056...e6a	Network activity	ip-dst	196.29.187.100			Mitm Node	✓	190	2	✓	Inherit	(0/0/0)		
2026-02-24	9aa...108	Network activity	ip-dst	188.226.192.35			Mitm Node	✓	190	2	✓	Inherit	(0/0/0)		
2026-02-24	4f1...390	Network activity	hostname	ns1.rootdnservers.com			Actor-controlled nameserver	✓	190	2	✓	Inherit	(0/0/0)		

IOC Analysis from MISP

(A) Infrastructure Types Used

Based on the MISP event and intelligence feed correlation, **Sea Turtle** uses a combination of the following infrastructure:

Command & Control (C2) Infrastructure

(1). Dedicated **operational nodes (C2 servers)** used for command execution, exfiltration, and control.

Examples from MISP intelligence:

- `185.64.105.100`
- `178.17.167.51`

Purpose:

- Malware control
- Session management
- Data exfiltration

- Remote access persistence

DNS & Network Manipulation Infrastructure

Primary Technique: DNS Hijacking

- Actor-controlled DNS infrastructure
- Compromised DNS registrars
- Rogue name servers
- MITM routing nodes

Examples from MISP:

- **Actor-controlled nameserver:**
  - `ns1.rootdnsservers.com`

MITM Nodes:

- `95.179.131.225`
- `140.82.58.253`

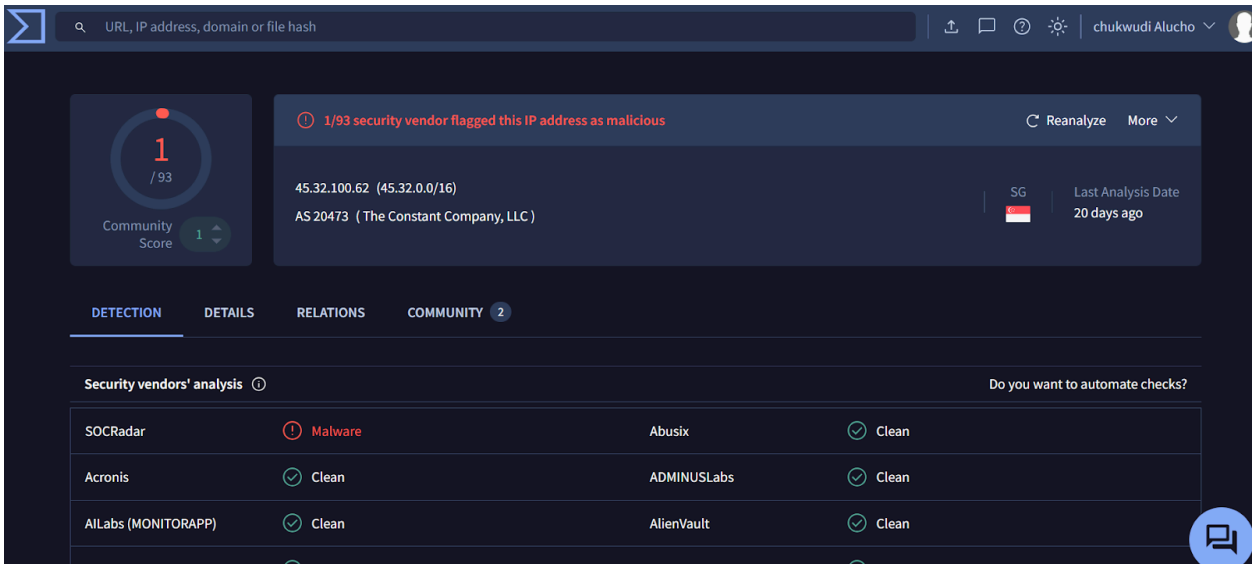
Purpose:

- Credential interception
- Session hijacking
- Traffic redirection
- Adversary-in-the-Middle (AiTM) attacks
- Credential harvesting

Infrastructure Type	Observed Usage by Sea Turtle	Assessment
Command & Control (C2) Servers	Confirmed	Primary operational infrastructure used for remote control and coordination.
DNS Hijacking Nodes	Confirmed	Core attack vector enabling traffic redirection and credential interception.
Rogue Nameservers	Confirmed	Used to manipulate DNS resolution and maintain persistence.
MITM Infrastructure	Confirmed	Supports Adversary-in-the-Middle operations and session interception.
Exploit Infrastructure	Confirmed	Used for compromising public-facing applications and network services.

Phishing Infrastructure	Limited / Secondary	Not primary tactic but may support credential harvesting operations.
Cloud Infrastructure Abuse	Possible	Potential use for staging, proxying, or obfuscating attack traffic.

OSINT Correlation Screen short from Virus **Total / AbuseIPBD**



AbuseIPDB » *193.46.255.33*

Check an IP Address, Domain Name, Subnet, or ASN
e.g. **24.159.130.130**, **microsoft.com**, **5.188.10.0/24**, or **AS15169**

24.159.130.130 CHECK

193.46.255.33 was found in our database!

This IP was reported **271,026** times. Confidence of Abuse is **100%**: ?

100%

ISP	UNMANAGED LTD
Usage Type	Data Center/Web Hosting/Transit
ASN	AS47890
Hostname(s)	hostingmailto181.statics.servermail.org

SPONSOR
Squarespace Squarespace makes marketing, customer management, and checkout effortless.

Mitre Att&ck Screen Short

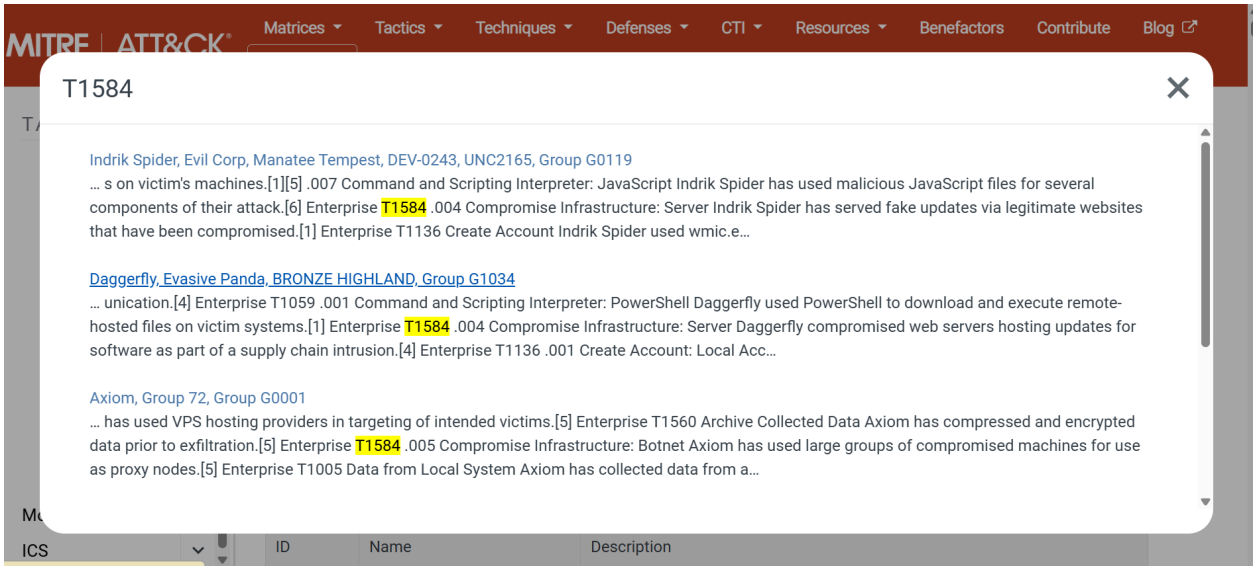
MITRE | ATT&CK® Matrices Tactics Techniques Defenses CTI Resources Benefactors Contribute Blog

T1557

LuminousMoth, Group G1014
… t, @KyawPyiytHtet; Zaw Min Htun, @Z3TAE Version: 1.0 Created: 23 February 2023 Last Modified: 16 April 2025 Version Permalink Live Version Techniques Used Domain ID Name Use Enterprise T1557 .002 Adversary-in-the-Middle: ARP Cache Poisoning LuminousMoth has used ARP spoofing to redirect a compromised machine to an actor-controlled website.[2] Enterprise T1071 .001 Application L…

Wizard Spider, UNC1878, TEMP.MixMaster, Grim Spider, FIN12, GOLD BLACKBURN, ITG23, Periwinkle Tempest, DEV-0193, Group G0102
… e use of net group "Domain admins" /DOMAIN. Wizard Spider has also leveraged the PowerShell cmdlet Get-ADComputer to collect account names from Active Directory data.[11][7] Enterprise T1557 .001 Adversary-in-the-Middle: LLMNR/NBT-NS Poisoning and SMB Relay Wizard Spider has used the Invoke-Inveigh PowerShell cmdlets, likely for name service poisoning.[4] Enterprise T1071 .001 …

Mustang Panda, TA416, RedDelta, BRONZE PRESIDENT, STATELY TAURUS, FIREANT, CAMARO DRAGON, EARTH PRETA, HIVE0154, TWILL TYPHOON, TANTALUM, LUMINOUS MOTH, UNC6384, TEMP.Hex, Red Lich, Group G0129
… at were re-registrations of expired domains.[32] .006 Acquire Infrastructure: Web Services Mustang Panda has set up Dropbox and Google Drive to host malicious downloads.[23] Enterprise T1557 Adversary-in-the-Middle Mustang Panda leveraged a captive portal hijack that redirected the victim to a webpage that prompted the victim to download a malicious payload.[30] Enterprise T107…



MISP Correlation Analysis

Correlations to Other Events

From my MISP screenshot:

Related Event Identified:

"Sea Turtle keeps on swimming, finds new victims, DNS hijacking techniques"
2019-07-09
Correlated Event Count: **15**

Meaning of Correlation:

- Indicates **campaign continuity**
- Shared IOCs across multiple MISP events
- Infrastructure reuse
- Recurrent targeting patterns
- Persistent threat actor operations

- Multi-year operational persistence

Intelligence Interpretation:

Sea Turtle is not an isolated campaign actor — it is a **persistent, long-term threat group** with:

- Reused infrastructure
- Repeated DNS-based attack chains
- Long operational timelines
- Strategic targeting

Threat Actor Profile – Sea Turtle

Known Aliases

- **Sea Turtle**
- **Teal Kurma**
- **Silicon**
- **Cosmic Wolf**

Country of Origin / Attribution

- **Suspected Origin:** Turkey
- **Attribution Type:** State-aligned / state-sponsored
- **Operational Alignment:** Turkish national strategic interests

Threat Motivation & Operational Intent Assessment

Category	Assessment	Analyst Evaluation
Espionage	Confirmed – Primary	Core operational objective focused on intelligence gathering and strategic data acquisition.
Intelligence Collection	Confirmed	Direct evidence of credential harvesting, network interception, and surveillance activity.

Primary Motivation:

Cyber espionage and intelligence collection

Common Tactics, Techniques, and Procedures (TTPs)

Primary Techniques

- **DNS Hijacking (Core Technique)**
- **Adversary-in-the-Middle (AiTM)**
- **MITM Attacks**
- **Digital Certificate Abuse**
- **Compromised Name Servers**
- **DNS Registrar Compromise**

Secondary Techniques

- Exploit Public-Facing Applications
- Credential Harvesting
- External Remote Services abuse
- Network traffic interception
- Infrastructure persistence

Mapped to MITRE ATT&CK:

-

ID	Technique Name	Sea Turtle Implementation
T1114.001	**Email Collection: Local Email Collection**	The group has been observed collecting email archives directly from victim environments (such as cPanel accounts) to gather political and economic intelligence.
T1070.002	**Indicator Removal: Clear Linux or Mac System Logs**	To evade detection, Sea Turtle frequently overwrites Linux system logs and unsets Bash history files to hide their post-exploitation commands.
T1588.004	**Obtain Capabilities: Digital Certificates**	A hallmark of this group is "certificate impersonation." They obtain CA-signed X.509 certificates for a target's domain from a different provider to facilitate Adversary-in-the-Middle (AiTM) attacks.

T1560.001	**Archive Collected Data: Archive via Utility**	Analysts have documented Sea Turtle using the standard tar utility to bundle collected email data and other sensitive files into archives before exfiltration.

ID	Technique Name	Sea Turtle Implementation
T1557	**Adversary-in-the-Middle (AiTM)**	**Core Strategy:** Sea Turtle is famous for DNS hijacking. By modifying DNS records at the registrar level, they position themselves as a "Man-in-the-Middle" to intercept credentials and traffic.
T1566	**Phishing**	Used for initial access. They often employ spear-phishing against employees of DNS registrars or telecommunications firms to steal the administrative credentials needed for their hijacking operations.
T1046	**Network Service Discovery**	Once they gain a foothold in a network (like a cPanel environment), they use scanning to identify internal services and lateral movement opportunities.

T1583	**Acquire Infrastructure**	They have been observed purchasing domains and renting VPS instances to host their "Shadow" name servers and command-and-control (C2) nodes.
T1584	**Compromise Infrastructure**	Instead of buying everything, they frequently compromise third-party DNS servers and legitimate web hosting environments to host their malicious tools.
T1071	**Application Layer Protocol**	Their custom malware (like **SnappyTCP**) typically uses standard **Web Protocols (HTTP/S)** for C2 communication to blend in with legitimate web traffic.

Sea Turtle TTP Mapping (Part 2)

ID	Technique Name	Sea Turtle Implementation
T1557	**Adversary-in-the-Middle (AiTM)**	**Core Strategy:** Sea Turtle is famous for DNS hijacking. By modifying DNS records at the registrar level, they position themselves as a "Man-in-the-Middle" to intercept credentials and traffic.

T1566	**Phishing**	Used for initial access. They often employ spear-phishing against employees of DNS registrars or telecommunications firms to steal the administrative credentials needed for their hijacking operations.
T1046	**Network Service Discovery**	Once they gain a foothold in a network (like a cPanel environment), they use scanning to identify internal services and lateral movement opportunities.
T1583	**Acquire Infrastructure**	They have been observed purchasing domains and renting VPS instances to host their "Shadow" name servers and command-and-control (C2) nodes.
T1584	**Compromise Infrastructure**	Instead of buying everything, they frequently compromise third-party DNS servers and legitimate web hosting

		environments to host their malicious tools.
T1071	**Application Layer Protocol**	Their custom malware (like **SnappyTCP**) typically uses standard **Web Protocols (HTTP/S)** for C2 communication to blend in with legitimate web traffic.

Analyst Note: The "Sea Turtle" Methodology

Sea Turtle doesn't just "hack a website"; they hack the **entire path** to that website.

1. **Preparation (T1583/T1584):** They set up or compromise infrastructure.
2. **Access (T1566/T1199):** They phish a service provider or exploit a trusted relationship.
3. **The Pivot (T1557):** They use their access to change DNS records (AiTM).
4. **The Payoff (T1071/T1114.001):** They intercept user logins and collect sensitive data/emails.

Notable Campaigns / Targets

Target Sectors:

- Government
- Military
- Intelligence agencies

- Diplomatic organizations
- Telecom providers
- National infrastructure

Geographic Focus:

- Middle East
- North Africa (MENA)
- Eastern Mediterranean

Countries Targeted:

- Turkey
- Greece
- Cyprus
- Iraq
- Syria
- Egypt

Tools / Malware / Capabilities

Tooling Types:

- Custom DNS manipulation toolkits
- Traffic redirection frameworks
- MITM proxy systems

- Credential harvesting platforms
- Network sniffing tools
- Infrastructure persistence scripts

Malware Class:

- Custom implants
- Network interception malware
- Proxy-based attack frameworks
- Credential harvesting payloads

4. Key IOCs Summary (from MISP Intelligence)

Infrastructure IOCs

Operational Nodes (C2):

- `185.64.105.100`
- `178.17.167.51`

MITM Nodes:

```
95.179.131.225
45.32.100.62

140.82.58.253
193.46.255.33
```

DNS Infrastructure:

- `ns1.rootdnsservers.com`

5. What Organizations Should Watch For

Detection Indicators

Network Indicators:

- Unauthorized DNS resolver changes
- Rogue name server entries
- Unexpected DNS propagation changes
- TLS certificate mismatches
- DNS registrar modification logs
- Traffic redirection anomalies
- MITM behavior patterns

Security Signals:

- Certificate chain anomalies
- Sudden domain resolution changes
- Unexpected IP routing paths
- Abnormal DNS TTL values
- Traffic flows through untrusted nodes

6. Executive Risk Summary

Sea Turtle represents a high-level strategic cyber threat characterized by:

- Long-term persistence
- State-aligned operations
- Infrastructure manipulation
- Intelligence-focused targeting
- Low-noise, high-impact operations

Business & National Risk:

- Data sovereignty compromise
- National security exposure
- Diplomatic data interception
- Regulatory compliance violations
- Long-term espionage exposure
- Supply-chain trust erosion

This actor does **not rely on noisy ransomware operations** — their threat model is **stealth, persistence, and long-term intelligence access**.

7. Mitigation Recommendations

Recommendation 1 – DNS & Infrastructure Hardening (Strategic)

Executive-Level Action:
Implement **DNS Security Governance** across the organization:

- DNSSEC enforcement
- Registrar account MFA
- Registry lock services
- DNS change monitoring
- Name server integrity validation
- Continuous DNS telemetry monitoring

Impact:
Prevents DNS hijacking, MITM attacks, and infrastructure compromise.

Recommendation 2 – Network Detection & Intelligence Integration (Operational)

SOC-Level Action:
Deploy **intelligence-driven detection engineering**:

- Suricata IDS for DNS anomaly detection
- MITM detection rules
- TLS certificate mismatch detection
- IOC-based alert correlation
- MISP → SIEM → IDS pipeline integration
- Automated threat feed ingestion

Impact:
Transforms threat intelligence into **real-time detection and prevention**.

Executive Conclusion

Sea Turtle is not a conventional cybercriminal group — it is a **strategic cyber espionage actor** using **infrastructure-level attacks** rather than malware-heavy operations. Their focus on DNS hijacking and MITM positioning allows them to silently intercept data at scale, making detection difficult and impact severe.

This intelligence demonstrates the importance of:

- **Proactive cyber defense**
- **Threat intelligence integration**
- **Infrastructure-layer security**
- **Detection engineering**
- **Continuous monitoring**

Final Leadership Message:

Organizations that rely only on endpoint security will not detect actors like Sea Turtle.
Defense must move to the network, infrastructure, and intelligence layers.